



GTC4Lusso USA - Axle shafts - Latest Update: 2021-01-26

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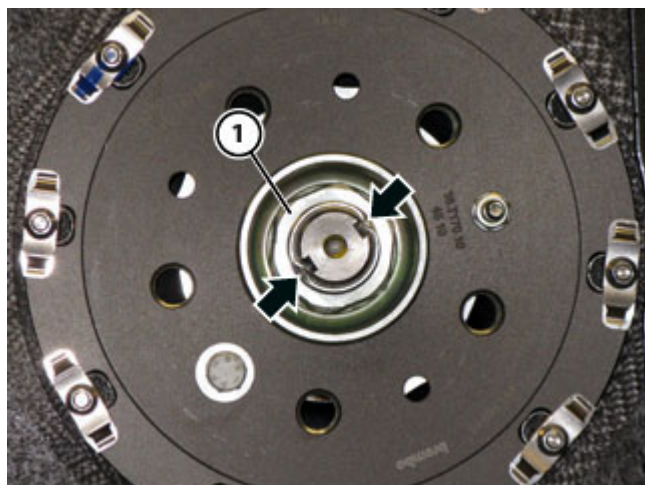
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C5.01 Axle shafts

Removing the front axle shafts

➤ Remove the complete wheels ( D2.01).

 Remove the front wheels.



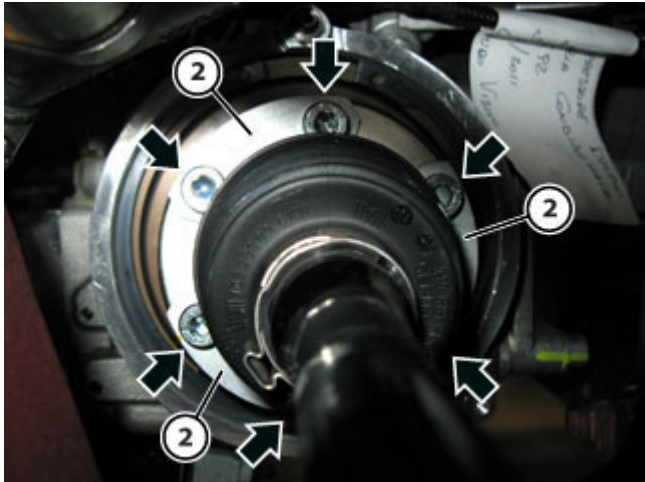
- Remove the indicated caulking using a punch and a hammer.
- Undo the nut (1).



- Undo the indicated nut and remove the lower arm pin from the hub carrier.



- Undo the indicated nut and remove the toe-in tie-rod pin from the hub carrier.



- Undo the screws indicated.
- Retrieve the shims (2).

- Tilt the hub carrier in order to detach the axle shaft from the PTU flange.
- Using a suitable punch and mallet, detach the axle shaft from the hub carrier bearing.
- Remove the axle shaft from the vehicle.

Refitting the front axle shafts

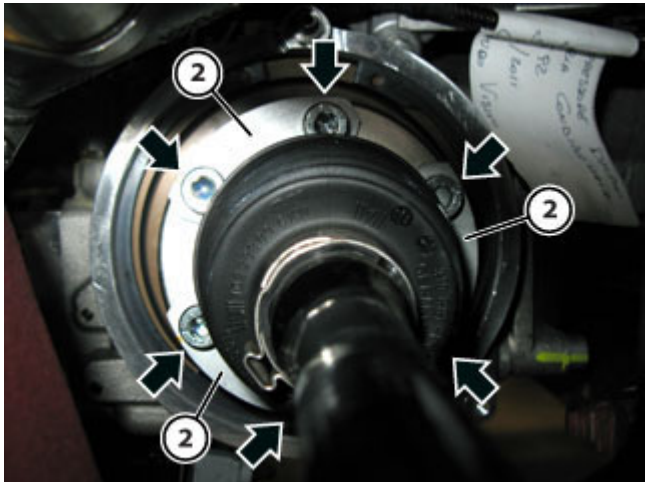
			
Tightening torque		Nm	Class
Fastening the front axle shafts	Screw	30 Nm	A
	Nut	275 Nm	A
Fastening the front suspension lower arms	Nut (pretightening)	90 Nm	A
	Nut	63 Nm	A
	Nut (pretightening)	9 Nm	A
	Nut	60 Nm	A
Fastening the steering tie-rods	Nut (pretightening)	40 Nm	A
	Nut	40 Nm	A

- Remove any residue of **Loctite® 638** from the splining on the axle shaft and on the bearing.



- Apply three even and regularly spaced circular beads of **Loctite® 638**.

- Fit the axle shaft in the hub carrier bearing.
- Check that the surface of the differential flange is completely clean.
- Tilt the hub carrier in order to fit the axle shaft onto the PTU flange.



- Fit the shims (2).
- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	30 Nm	A



- Fit the lower arm pin on the hub carrier and pretighten the indicated nut.

		
Tightening torque	Nm	Class
Nut (pretightening)	90 Nm	A

- Loosen and tighten the indicated nut.

		
Tightening torque	Nm	Class
Nut	63 Nm	A

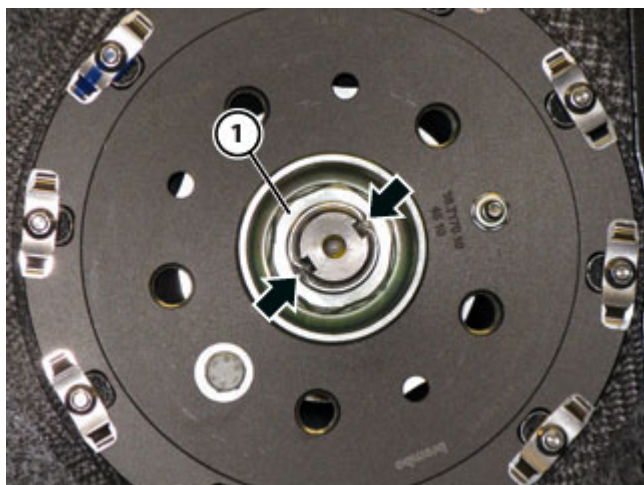


- Fit the toe-in tie-rod pin on the hub carrier and pre-tighten the indicated nut.

		
Tightening torque	Nm	Class
Nut (pretightening)	40 Nm	A

- Loosen and tighten the indicated nut.

		
Tightening torque	Nm	Class
Nut	40 Nm	A



- Tighten the nut (1).

		
Tightening torque	Nm	Class
Nut	275 Nm	A

- Loosen the nut (1) then tighten again.

		
Tightening torque	Nm	Class
Nut	275 Nm	A

- Caulk as indicated with a punch and a hammer.

➤ Refit the complete wheels ( D2.01).

 Refit the front wheels.

Removing the rear axle shafts

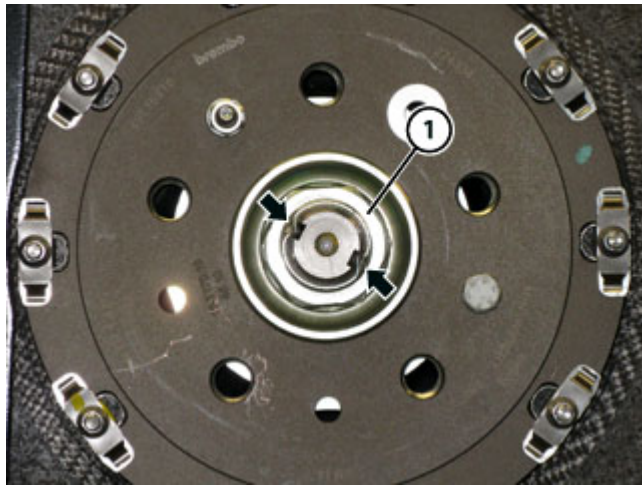
➤ Disable the EPB electric parking brake Autohold function ( D3.06).

➤ Remove the complete wheels ( D2.01).

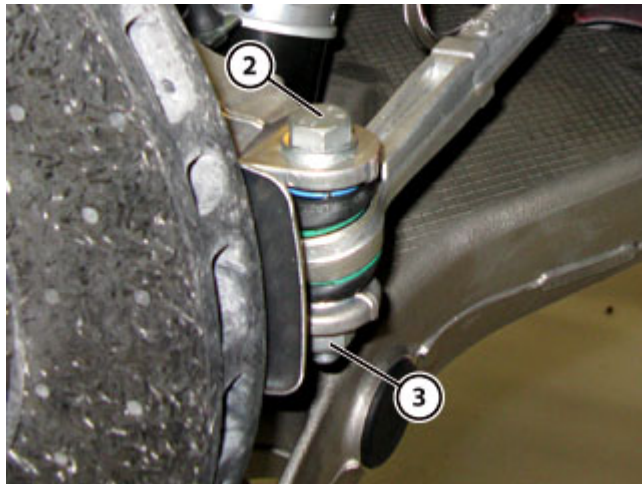
 Remove the rear wheels.

➤ Remove the flat undertray sections ( E3.13).

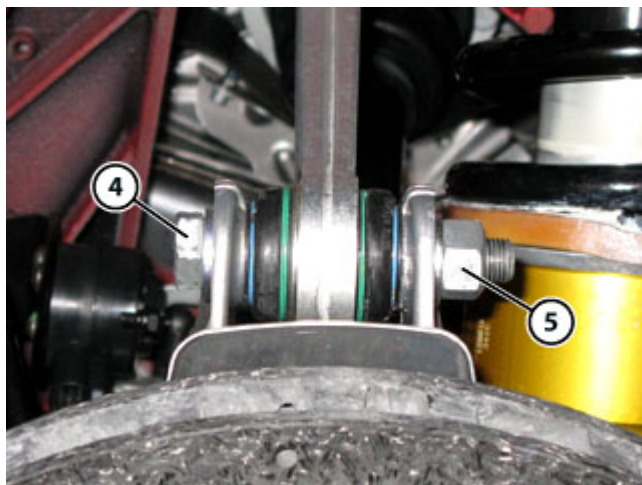
 Remove the rear flat undertray section.



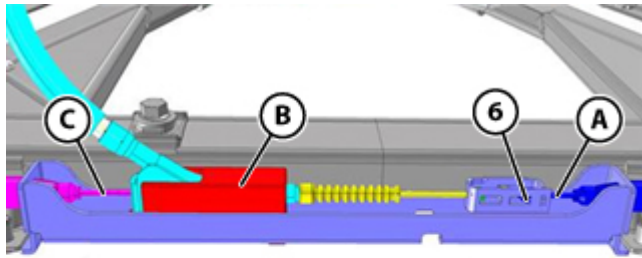
- Remove the indicated caulking using a punch and a hammer.
- Undo the nut (1).



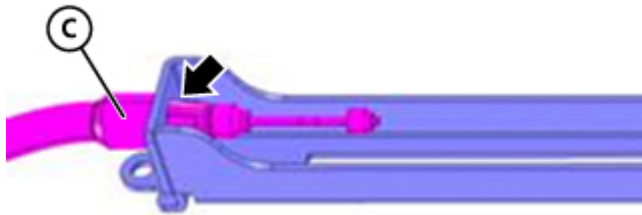
- Undo the nut (3), while holding the pin (2) to prevent it from turning.
- Remove the pin (2).
- Detach the toe-in tie-rod from the hub carrier.



- Undo the nut (5), while holding the pin (4) to prevent it from turning.
- Remove the pin (4).
- Detach the upper arm from the hub carrier.



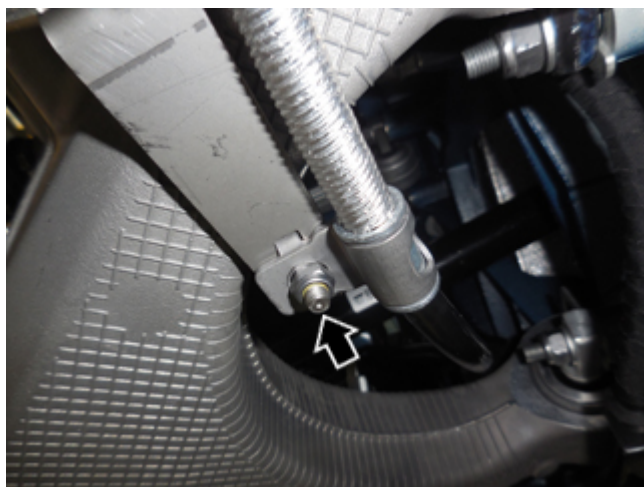
- Detach and remove the clip **(6)**.
- Detach the end of the right hand EPB Bowden cable **(A)**.
- Move the EPB equaliser **(B)** to the left.
- Detach the end of the left hand EPB Bowden cable **(C)** from the EPB equaliser **(B)**.



- Detach the sheath of the EPB Bowden cable **(C)** from the bracket.



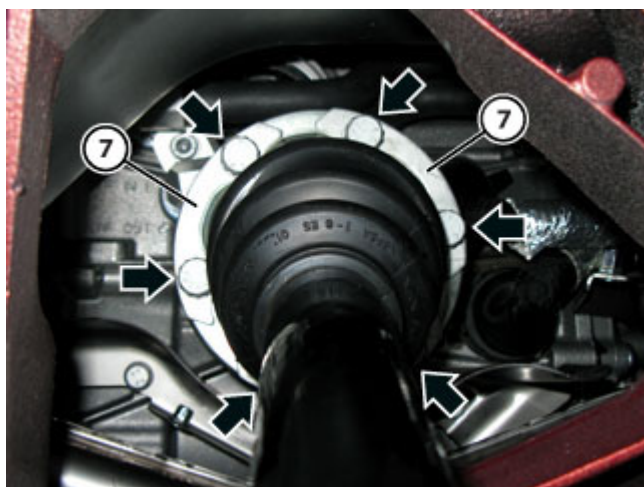
- Undo the indicated nut.



- Undo the indicated nut.



- Undo the indicated nut.



- Undo the screws indicated.
- Retrieve the three shims (7).

- Tilt the hub carrier to detach the axle shaft from the differential flange.
- Using a suitable punch and mallet, detach the axle shaft from the hub carrier bearing.
- Remove the axle shaft from the vehicle.

Refitting the rear axle shafts



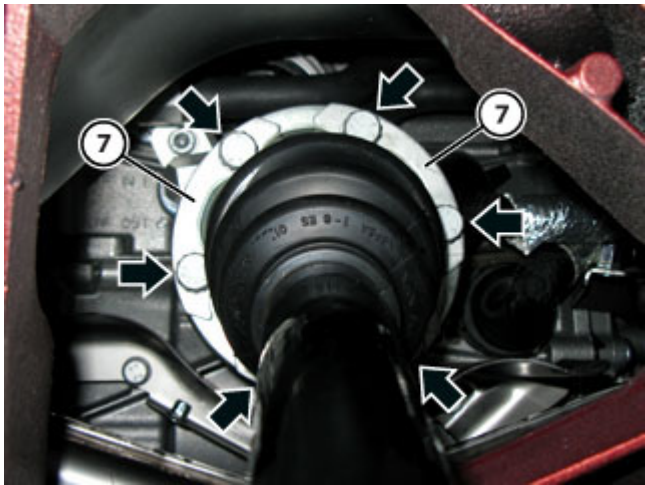
ng torque	Nm	Class

Tightening torque		Nm	Class
Fastening the rear axle shafts	Screw	78 Nm	A
	Screw (pretightening)	30 Nm	B
	Nut	275 Nm	A
Fastening EPB Bowden cables	Nut	9 Nm	B
	Screw	8 Nm	B
	Nut	7 Nm	B
	Nut	5 Nm	B
	Screw	9 Nm	B
Fastening the rear suspension upper arms	Nut	90 Nm	A
	Nut (pretightening)	9 Nm	A
	Nut	60 Nm	A
Fastening the rear toe-in tie-rods	Nut	80 Nm	A
	Nut (pretightening)	9 Nm	A
	Nut	25 Nm	A

- Remove any residue of **Loctite® 638** from the splining on the axle shaft and on the bearing.
- Apply three even and regularly spaced circular beads of **Loctite® 638**.



- Fit the axle shaft in the hub carrier bearing.
- Check that the surface of the differential flange is completely clean.
- Tilt the hub carrier in order to fit the axle shaft onto the differential flange.



- Fit the three shims (7).
- Pre-tighten the indicated screws.

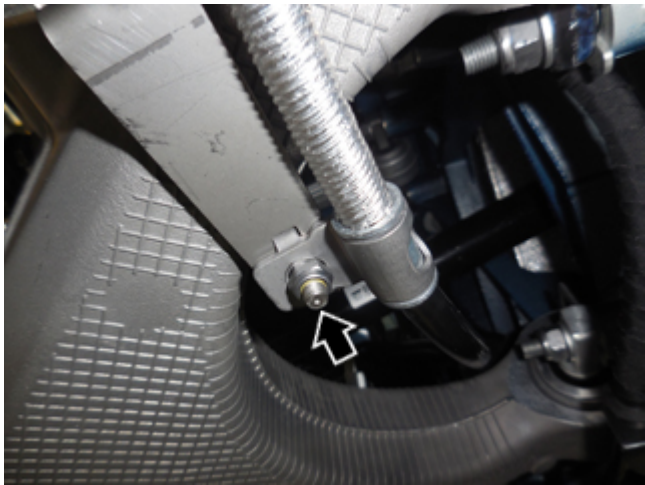
		
Tightening torque	Nm	Class
Screw (pretightening)	30 Nm	B

- Tighten the screws indicated.

		
Tightening torque	Nm	Class
Screw	78 Nm	A



- Tighten the indicated nut.



- Tighten the indicated nut.

		
Tightening torque	Nm	Class
Nut	5 Nm	B

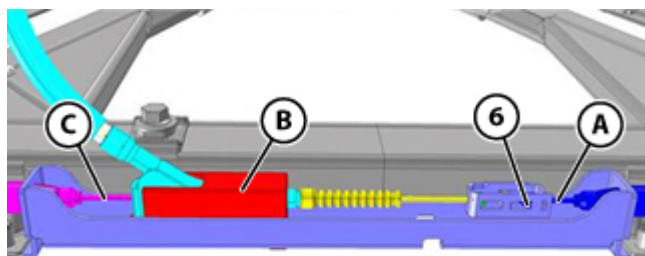
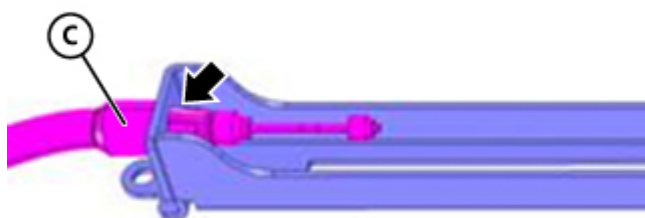


- Tighten the indicated nut.

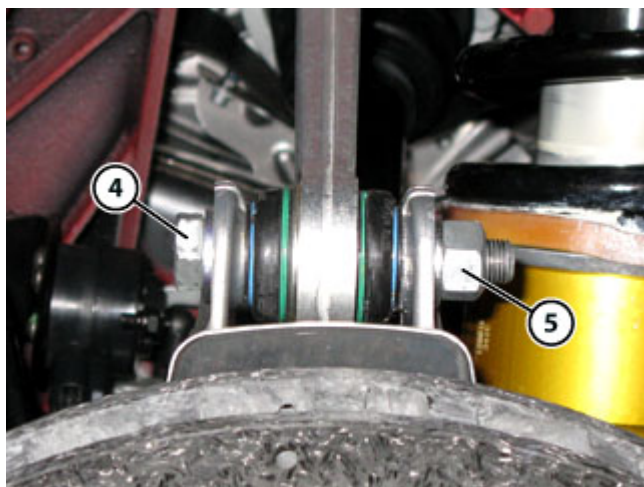


Tightening torque	Nm	Class
Nut	7 Nm	B

- Attach the sheath of the EPB Bowden cable (C) to the bracket.

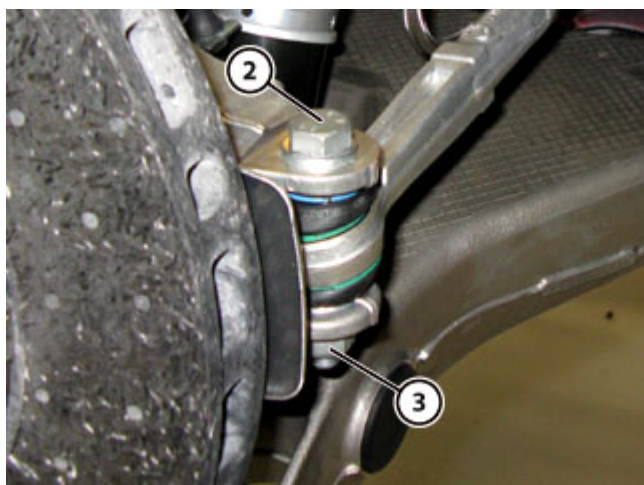


- Attach the end of the left hand EPB Bowden cable (C) onto the EPB equaliser (B).
- Move the EPB equaliser (B) to the right.
- Attach the end of the right hand EPB Bowden cable (A).
- Fit and attach the clip (6).



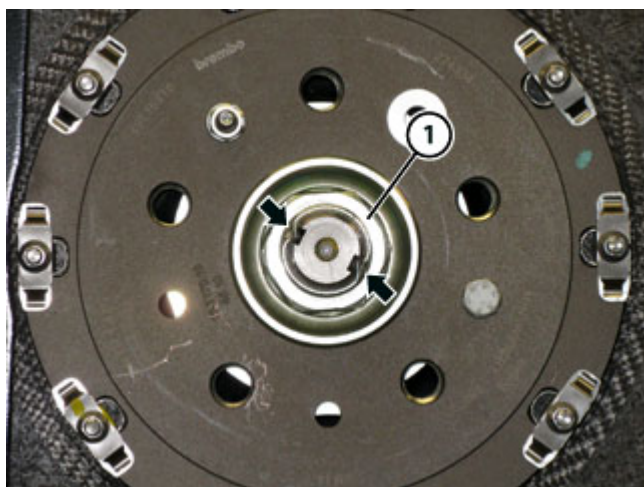
- Fit the upper arm on the hub carrier.
- Install the pin (4).
- Tighten the nut (5), while holding the pin (4) to prevent it from turning.

		
Tightening torque	Nm	Class
Nut	90 Nm	A



- Fit the toe-in tie-rod on the hub carrier.
- Install the pin (2).
- Tighten the nut (3), while holding the pin (2) to prevent it from turning.

		
Tightening torque	Nm	Class
Nut	80 Nm	A



- Tighten the nut (1).

		
Tightening torque	Nm	Class
Nut	275 Nm	A

- Loosen the nut (1) then tighten again.

		
Tightening torque	Nm	Class
Nut	275 Nm	A

- Caulk as indicated with a punch and a hammer.

➤ Refit the flat undertray sections ( E3.13).

 Refit the rear flat undertray section.

➤ Refit the complete wheels ( D2.01).

 Refit the rear wheels.

➤ Calibrate the EPB electric parking brake ( D3.06).

 This procedure must only be performed if, at Key-ON, the NPB parking brake node identifies a fault, indicated on the instrument panel and on the TFT display by the specific warning lamps, and with error **C155A** on the **NPB (EPB calibration error)** and error **C1313** on the **NFR (parking brake error)**.